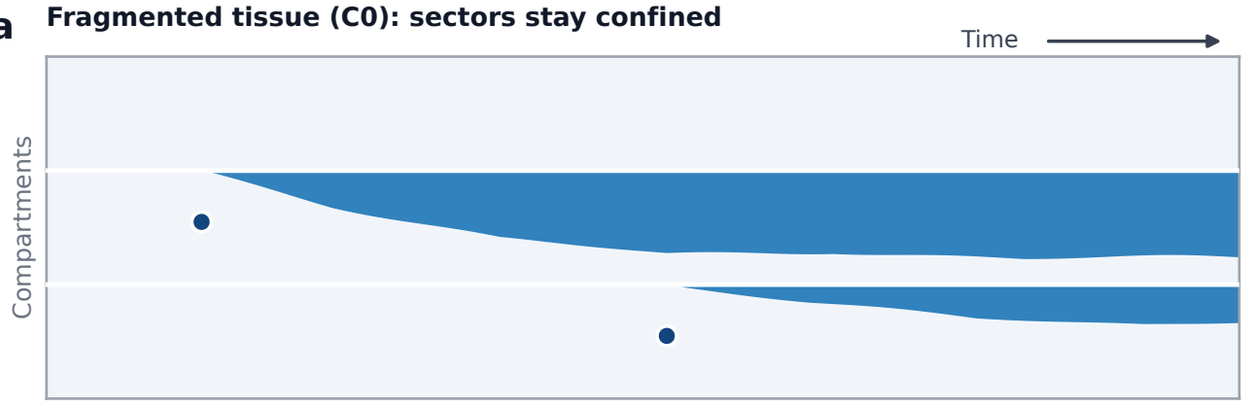
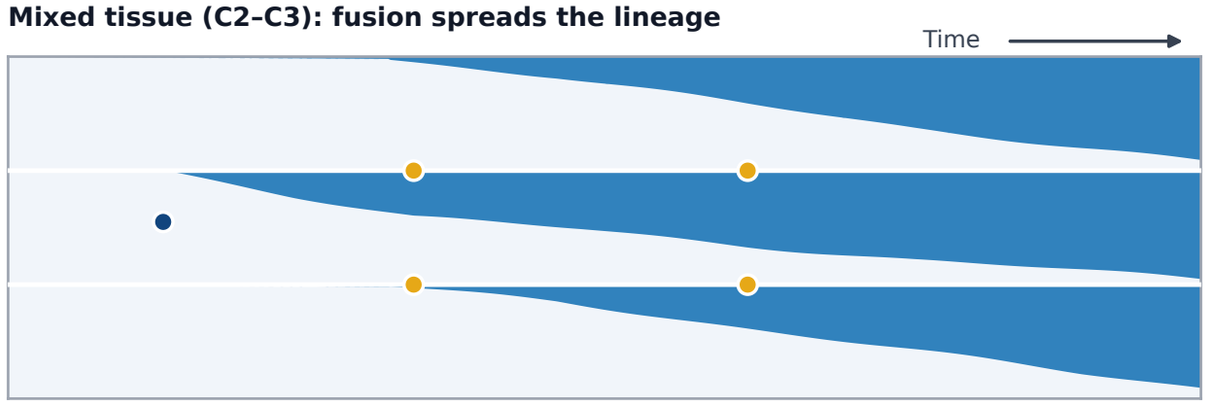
a

Fragmented tissue (C0): sectors stay confined



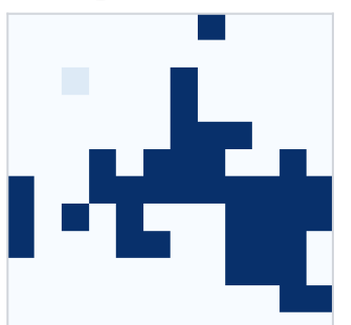
Mixed tissue (C2–C3): fusion spreads the lineage

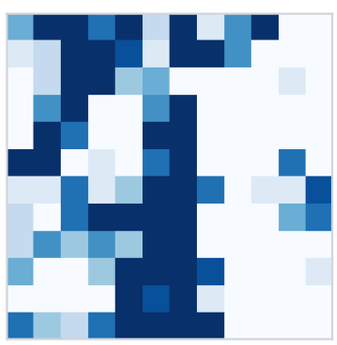


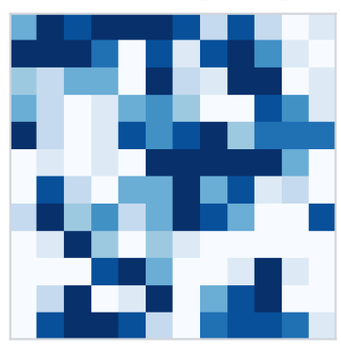
wild-type shading · core-gene-loss shading · core-gene loss (mutation, μ) · nuclear migration (hyphal fusion)

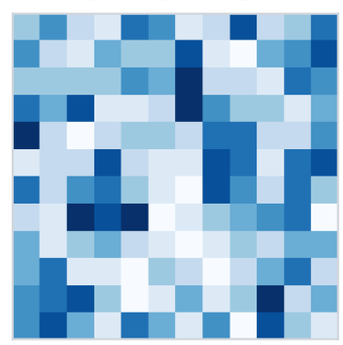
b

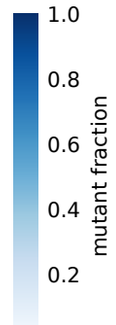
Fragmented (C0) | **Local fusion (C1)** | **Local + rare long-range (C2)** | **Strong long-range (C3)**



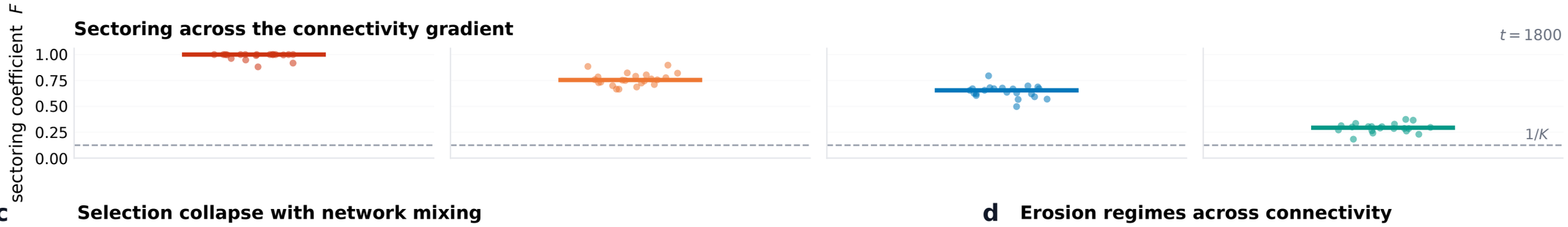






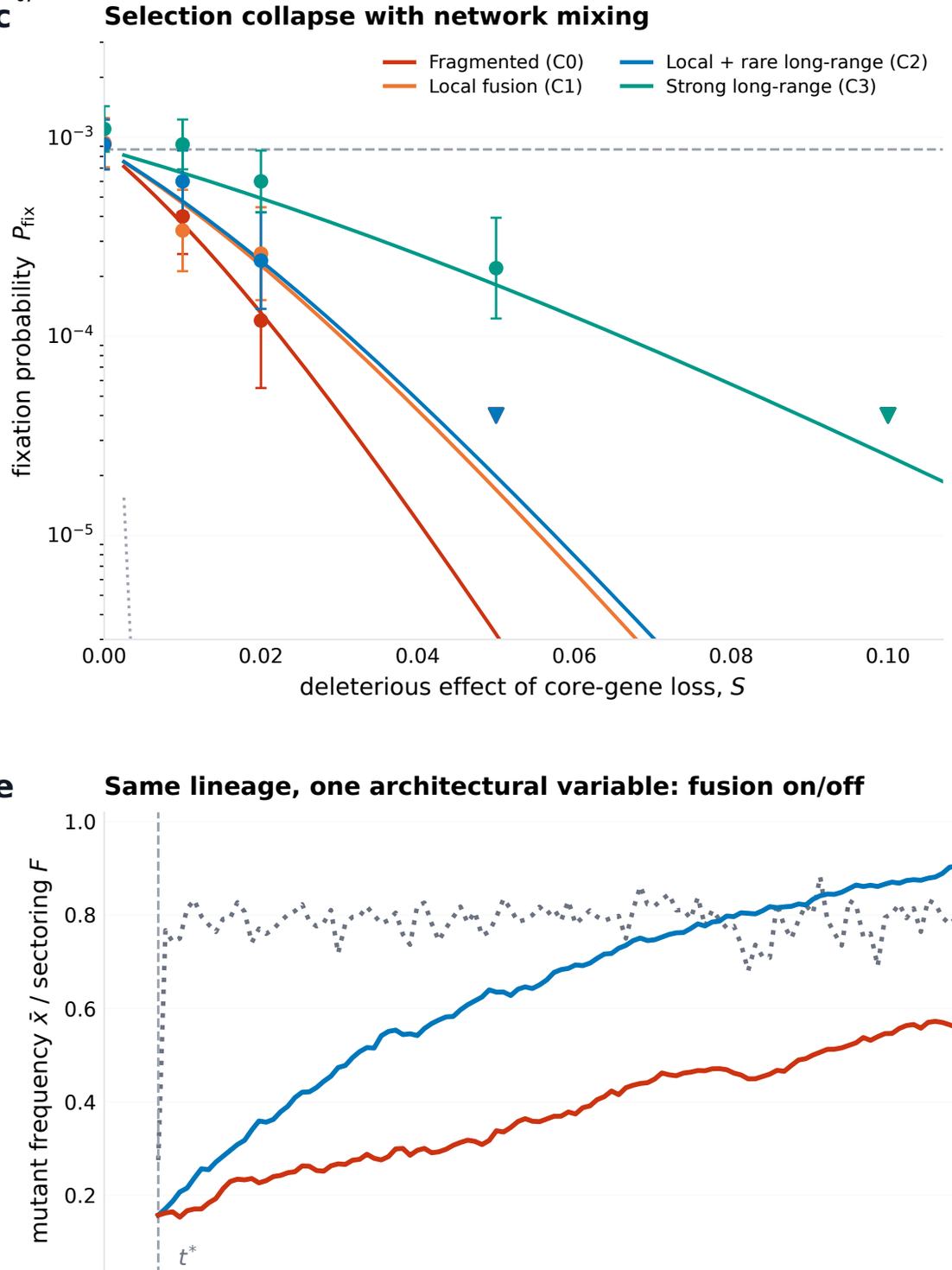


Sectoring across the connectivity gradient



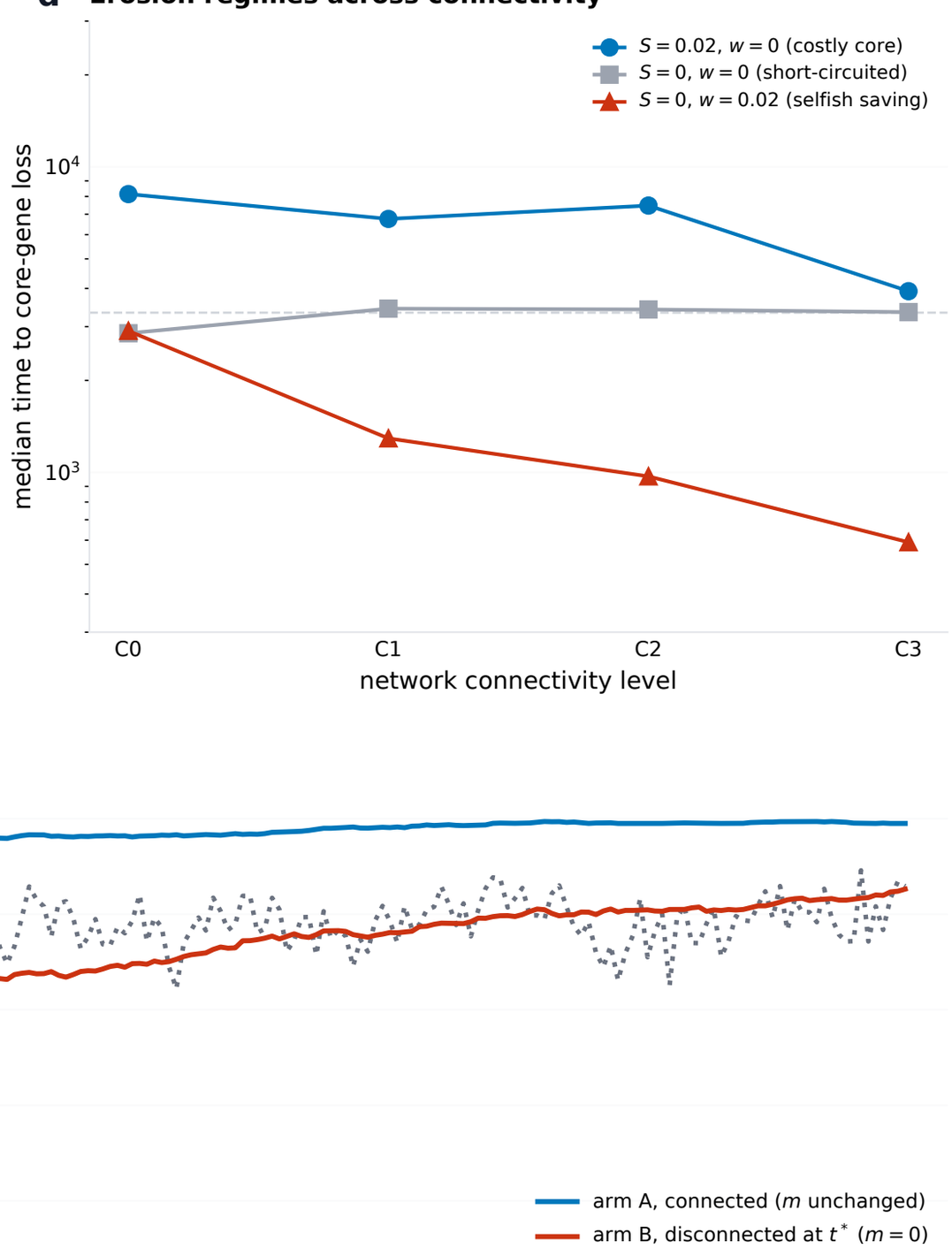
c

Selection collapse with network mixing



d

Erosion regimes across connectivity



e

Same lineage, one architectural variable: fusion on/off

